

SIT292 Module 7 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score
Question 1	1/1
Question 2	1/1
Question 3	2/2
Question 4	2/2
Question 5	3/3
Question 6	1/1
Total	10/10 (100%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 7 Self-Assessment
Session ID:	1042648011765624
Exam Start:	Fri Sep 11 2026 21:33:26
Exam Stop:	Fri Sep 11 2026 22:06:41
Time Spent:	00:33:15

Question 1

The system of equations,

$$\begin{aligned} -2x - 4y &= -5 \\ -6x + 5y &= -2 \\ -2x + y &= 3 \end{aligned}$$

where $x, y \in \mathbb{R}$, has no solution.

Find the vector $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \underline{0.774} \\ \underline{0.752} \end{pmatrix} \quad \checkmark$$

Expected answer: $\begin{pmatrix} \underline{41/53} \\ \underline{239/318} \end{pmatrix}$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

✓ Your answer is correct. You were awarded 1 mark.

Score: 1/1 ✓

Question 2

Find the least squares line of best fit $y = a + bx$ for the data points:

$$\{(6, 4), (6, -2), (4, 5), (-4, -6)\}$$

$$a = \underline{-2.088} \quad \checkmark$$

Expected answer: -2.0882352941

$$b = \underline{0.779} \quad \checkmark$$

Expected answer: 0.7794117647

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Gap 0

Score: 1/1

- Your answer is correct. You were awarded **0.5** marks.

Gap 1

- Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

Question 3

Find a basis and calculate the dimension for the following subspace of 2×2 matrices (with standard addition and multiplication by a scalar operations)

$$\text{span} \left\{ \begin{pmatrix} -3 & 9 \\ -3 & -13 \end{pmatrix}, \begin{pmatrix} 1 & -2 \\ -2 & 0 \end{pmatrix}, \begin{pmatrix} -2 & 5 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} -3 & 4 \\ 10 & 4 \end{pmatrix} \right\}$$

Enter up to 4 basis matrices. For any matrices not required in the basis, enter

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

$$\begin{pmatrix} \frac{-3}{-3} & \frac{9}{-13} \end{pmatrix}$$

Expected answer: $\begin{pmatrix} \frac{1}{-2} & \frac{-2}{0} \end{pmatrix},$

$$\begin{pmatrix} \frac{1}{-2} & \frac{-2}{0} \end{pmatrix}$$

Expected answer: $\begin{pmatrix} \frac{0}{-2} & \frac{1}{-2} \end{pmatrix},$

$$\begin{pmatrix} \frac{-2}{3} & \frac{5}{0} \end{pmatrix}$$

Expected answer: $\begin{pmatrix} \frac{0}{1} & \frac{0}{2} \end{pmatrix},$

$$\begin{pmatrix} \frac{-3}{10} & \frac{4}{4} \end{pmatrix}$$

Expected answer: $\begin{pmatrix} \frac{0}{0} & \frac{0}{1} \end{pmatrix}$

What is the dimension of the subspace?

4

Expected answer: 4

✓ Correct. Your basis matrices are independent and span the entire subspace. You were awarded 2 marks.

Score: 2/2 ✓

Question 4

The vector space spanned by the following set of polynomials is a subspace of polynomials of degree $n \leq 3$.

$$\{3 + 6x + 2x^2 - 10x^3, \quad -2 - 4x - 2x^2 + 10x^3, \quad -3 - 6x - 3x^2 + 9x^3, \quad -2 -$$

Identify a basis of the subspace. Enter the coefficients as rows of a matrix such that

Rows: 3 Columns: 4

$$\begin{pmatrix} \underline{3} & \underline{6} & \underline{2} & \underline{-10} \\ \underline{-3} & \underline{-6} & \underline{-3} & \underline{9} \\ \underline{-2} & \underline{-4} & \underline{0} & \underline{4} \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} \underline{1} & \underline{2} & \underline{0} & \underline{-2} \\ \underline{0} & \underline{0} & \underline{1} & \underline{-2} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ x \\ x^2 \\ x^3 \end{pmatrix} = \left(\text{basis polynomials} \right)$$

(reduce the size of the matrix if the dimension is less than 4)

What is the dimension of the subspace?

3

Expected answer: 3

✓ Correct. Your basis polynomials are independent and span the entire subspace. You were awarded 2 marks.

Score: 2/2 ✓

Question 5

The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & -2 & -2 & 1 & 1 \\ 2 & -4 & -4 & -2 & 1 \\ 2 & -4 & -4 & 2 & 0 \\ 2 & -4 & -4 & -2 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & -2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Let W be a subspace of $\mathbf{P}_3(\mathbb{R})$ spanned by the set

$$S = \{1 + 2x + 2x^2 + 2x^3, -2 - 4x - 4x^2 - 4x^3, -2 - 4x - 4x^2 - 4x^3\}$$

a)

Decide whether or not S is linearly independent. If **not**, write the **second** polynomial as a linear combination of the others. If it *is* linearly independent, write 0 for both coefficients.

$$-2 - 4x - 4x^2 - 4x^3 = \underline{-2} \quad \checkmark$$

polynomial 2

Expected answer: -2

$$(1 + 2x + 2x^2 + 2x^3) + \underline{0} \quad \checkmark$$

polynomial 1

Expected answer: 0

$$(-2 - 4x - 4x^2 - 4x^3)$$

polynomial 3

▼ Feedback

Score: 1/1 ✓

- ✓ Your coefficients are correct. You were awarded 1 mark.

b)

(i) Find a subset of S that is a basis for W . (Select the polynomials to include in the subset)

☒ $1 + 2x + 2x^2 + 2x^3$ ☐ $-2 - 4x - 4x^2 - 4x^3$

☐ $-2 - 4x - 4x^2 - 4x^3$

Expected answer:

☒ $1 + 2x + 2x^2 + 2x^3$ ☐ $-2 - 4x - 4x^2 - 4x^3$

☐ $-2 - 4x - 4x^2 - 4x^3$

(ii) Hence find the dimension of W : 1

Expected answer: 1

▼ Feedback

Score: 1/1 ✓

- ✓ Correct. Dimension is 1 and so any of the polynomials in S can be used as a basis. You were awarded 1 mark.

c)

Find a basis for the **row space** of A . Enter your basis as a matrix comprised of your basis row vectors. You can increase the number of rows as required.

Rows: 3 Columns: 5

$$\begin{pmatrix} \underline{1} & \underline{-2} & \underline{-2} & \underline{0} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} \underline{1} & \underline{-2} & \underline{-2} & \underline{0} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} \end{pmatrix}$$

▼ Feedback

Score: 1/1 ✓

- ✓ Basis given for the row space of A is correct. You were awarded 1 mark.

Question 6

The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & -1 & -2 & 1 & 1 \\ 2 & -2 & -4 & -2 & 1 \\ 2 & -2 & -4 & 1 & -1 \\ 1 & -1 & -2 & -1 & -1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Select a subset of matrices from the following which can be used to give a basis that spans $\mathbf{M}_{2,2}(\mathbb{R})$?

☐ $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$
☐ $\begin{pmatrix} -1 & -2 \\ -2 & -1 \end{pmatrix}$
☐ $\begin{pmatrix} -2 & -4 \\ -4 & -2 \end{pmatrix}$
☐ $\begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$
☒ Not possible

Expected answer:

☐ $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$
☐ $\begin{pmatrix} -1 & -2 \\ -2 & -1 \end{pmatrix}$
☐ $\begin{pmatrix} -2 & -4 \\ -4 & -2 \end{pmatrix}$
☐ $\begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$
☒ Not possible

✓ Correct. It is not possible to form a basis that spans the set of real 2x2 matrices with the elements given. You were awarded 1 mark.

Score: 1/1 ✓

Module 7 Self-Assessment

Working & Annotations

Detailed mathematical working supporting the completed Numbas self-assessment questions

Question 1

Question.

The system

$$\begin{aligned} -2x - 4y &= -5, \\ -6x + 5y &= -2, \\ -2x + y &= 3 \end{aligned}$$

has no solution. Find the vector

$$z = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

that best approximates a solution.

Concepts / properties applied:

Matrix representation of an inconsistent system; transpose; normal equations; Theorem 5.6.1; Gaussian elimination.

Method.

Write the system as $Az = b$, form

$$(A^T A)z = A^T b,$$

and solve the resulting 2×2 system.

Step-by-step working

The coefficient matrix and right-hand side are

$$A = \begin{bmatrix} -2 & -4 \\ -6 & 5 \\ -2 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} -5 \\ -2 \\ 3 \end{bmatrix}.$$

Let

$$z = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}.$$

The normal equations are

$$(A^T A)z = A^T b. \quad (\text{Theorem 5.6.1})$$

First,

$$A^T = \begin{bmatrix} -2 & -6 & -2 \\ -4 & 5 & 1 \end{bmatrix}.$$

Hence,

$$\begin{aligned}
 A^T A &= \begin{bmatrix} -2 & -6 & -2 \\ -4 & 5 & 1 \end{bmatrix} \begin{bmatrix} -2 & -4 \\ -6 & 5 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 4 + 36 + 4 & 8 - 30 - 2 \\ 8 - 30 - 2 & 16 + 25 + 1 \end{bmatrix} \\
 &= \begin{bmatrix} 44 & -24 \\ -24 & 42 \end{bmatrix}. \quad (\text{Matrix multiplication})
 \end{aligned}$$

Also,

$$A^T b = \begin{bmatrix} -2 & -6 & -2 \\ -4 & 5 & 1 \end{bmatrix} \begin{bmatrix} -5 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 + 12 - 6 \\ 20 - 10 + 3 \end{bmatrix} = \begin{bmatrix} 16 \\ 13 \end{bmatrix}. \quad (\text{Matrix multiplication})$$

Therefore,

$$\begin{bmatrix} 44 & -24 \\ -24 & 42 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} 16 \\ 13 \end{bmatrix},$$

so

$$44x_0 - 24y_0 = 16,$$

$$-24x_0 + 42y_0 = 13.$$

Divide the first equation by 4:

$$11x_0 - 6y_0 = 4.$$

Divide the second equation by 3:

$$-8x_0 + 14y_0 = \frac{13}{3}.$$

Multiply the first simplified equation by 7:

$$77x_0 - 42y_0 = 28.$$

Multiply the second simplified equation by 3:

$$-24x_0 + 42y_0 = 13.$$

Adding,

$$53x_0 = 41,$$

hence

$$\boxed{x_0 = \frac{41}{53}}.$$

Substituting into

$$11x_0 - 6y_0 = 4,$$

$$11 \left(\frac{41}{53} \right) - 6y_0 = 4,$$

so

$$6y_0 = \frac{451}{53} - \frac{212}{53} = \frac{239}{53}.$$

Therefore,

$$y_0 = \frac{239}{318}.$$

Verification

Numerically,

$$x_0 = \frac{41}{53} \approx 0.773585, \quad y_0 = \frac{239}{318} \approx 0.751572.$$

Thus

$$z \approx \begin{bmatrix} 0.774 \\ 0.752 \end{bmatrix}.$$

These are the values entered in Numbas and were accepted as correct.

Final answer

$$z = \begin{bmatrix} \frac{41}{53} \\ \frac{239}{318} \end{bmatrix} \approx \begin{bmatrix} 0.774 \\ 0.752 \end{bmatrix}$$

Key Learning Point.

For an inconsistent system, the normal equations convert the original problem into a consistent system whose solution gives the least-squares best approximation.

Question 2

Question.

Find the least-squares line of best fit

$$y = a + bx$$

for the data points

$$\{(6, 4), (6, -2), (4, 5), (-4, -6)\}.$$

Concepts / properties applied:

Least-squares line; Theorem 5.6.2; normal equations; matrix multiplication; Gaussian elimination.

Method.

Write the line as

$$Mz = y, \quad z = \begin{bmatrix} a \\ b \end{bmatrix},$$

then solve

$$(M^T M)z = M^T y.$$

Step-by-step working

The data give

$$M = \begin{bmatrix} 1 & 6 \\ 1 & 6 \\ 1 & 4 \\ 1 & -4 \end{bmatrix}, \quad y = \begin{bmatrix} 4 \\ -2 \\ 5 \\ -6 \end{bmatrix}.$$

Let

$$z = \begin{bmatrix} a \\ b \end{bmatrix}.$$

The normal equations are

$$(M^T M)z = M^T y. \quad (\text{Theorem 5.6.2})$$

First,

$$M^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 6 & 6 & 4 & -4 \end{bmatrix}.$$

Hence,

$$\begin{aligned} M^T M &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 6 & 6 & 4 & -4 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 1 & 6 \\ 1 & 4 \\ 1 & -4 \end{bmatrix} = \begin{bmatrix} 4 & 12 \\ 12 & 36 + 36 + 16 + 16 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 12 \\ 12 & 104 \end{bmatrix}. \quad (\text{Matrix multiplication}) \end{aligned}$$

Also,

$$\begin{aligned} M^T y &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 6 & 6 & 4 & -4 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 5 \\ -6 \end{bmatrix} = \begin{bmatrix} 4 - 2 + 5 - 6 \\ 24 - 12 + 20 + 24 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 56 \end{bmatrix}. \quad (\text{Matrix multiplication}) \end{aligned}$$

Therefore,

$$\begin{bmatrix} 4 & 12 \\ 12 & 104 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 56 \end{bmatrix}.$$

Hence,

$$4a + 12b = 1,$$

$$12a + 104b = 56.$$

Divide the first equation by 4:

$$a + 3b = \frac{1}{4}.$$

Thus,

$$a = \frac{1}{4} - 3b.$$

Substitute into the second equation:

$$12 \left(\frac{1}{4} - 3b \right) + 104b = 56.$$

Therefore,

$$3 - 36b + 104b = 56,$$

so

$$68b = 53.$$

Hence,

$$\boxed{b = \frac{53}{68}}.$$

Now,

$$a = \frac{1}{4} - 3 \left(\frac{53}{68} \right) = \frac{17}{68} - \frac{159}{68} = -\frac{142}{68} = -\frac{71}{34}.$$

Thus,

$$a = -\frac{71}{34}.$$

Verification

Numerically,

$$a = -\frac{71}{34} \approx -2.088235,$$

$$b = \frac{53}{68} \approx 0.779412.$$

Therefore the fitted line is

$$y \approx -2.088 + 0.779x.$$

The values $a = -2.088$ and $b = 0.779$ were accepted by Numbas.

Final answer

$$a = -\frac{71}{34} \approx -2.088, \quad b = \frac{53}{68} \approx 0.779$$

Key Learning Point.

A least-squares line is obtained by treating the intercept and slope as unknown coefficients and solving the corresponding normal equations.

Question 3

Question.

Find a basis and calculate the dimension for the following subspace of 2×2 matrices:

$$\text{span} \left\{ \begin{bmatrix} -3 & 9 \\ -3 & -13 \end{bmatrix}, \begin{bmatrix} 1 & -2 \\ -2 & 0 \end{bmatrix}, \begin{bmatrix} -2 & 5 \\ 3 & 0 \end{bmatrix}, \begin{bmatrix} -3 & 4 \\ 10 & 4 \end{bmatrix} \right\}.$$

Concepts / properties applied:

Matrix-vector correspondence; linear independence; determinant; basis; dimension.

Method.

Represent each 2×2 matrix by a vector in $\mathbb{R}^4$, then test the four resulting vectors for linear independence.

Step-by-step working

Let

$$M_1 = \begin{bmatrix} -3 & 9 \\ -3 & -13 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 1 & -2 \\ -2 & 0 \end{bmatrix},$$

$$M_3 = \begin{bmatrix} -2 & 5 \\ 3 & 0 \end{bmatrix}, \quad M_4 = \begin{bmatrix} -3 & 4 \\ 10 & 4 \end{bmatrix}.$$

Represent these matrices by their entries in the order

$$(a_{11}, a_{12}, a_{21}, a_{22}).$$

Thus,

$$v_1 = \begin{bmatrix} -3 \\ 9 \\ -3 \\ -13 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -2 \\ -2 \\ 0 \end{bmatrix},$$

$$v_3 = \begin{bmatrix} -2 \\ 5 \\ 3 \\ 0 \end{bmatrix}, \quad v_4 = \begin{bmatrix} -3 \\ 4 \\ 10 \\ 4 \end{bmatrix}.$$

Place these vectors as columns:

$$C = \begin{bmatrix} -3 & 1 & -2 & -3 \\ 9 & -2 & 5 & 4 \\ -3 & -2 & 3 & 10 \\ -13 & 0 & 0 & 4 \end{bmatrix}.$$

To test linear independence, calculate

$$\det(C) = 2 \neq 0. \quad (\text{Determinant})$$

Therefore,

$$\boxed{\text{rank}(C) = 4}.$$

Hence the four matrices are linearly independent.

Since there are four independent matrices in a 4-dimensional matrix space,

$$\boxed{\{M_1, M_2, M_3, M_4\}}$$

is a basis for the subspace.

Verification

The determinant is non-zero, so no non-trivial linear combination of the four matrices can produce the zero matrix.

Thus all four original matrices are needed.

Final answer

$$\mathcal{B} = \left\{ \begin{bmatrix} -3 & 9 \\ -3 & -13 \end{bmatrix}, \begin{bmatrix} 1 & -2 \\ -2 & 0 \end{bmatrix}, \begin{bmatrix} -2 & 5 \\ 3 & 0 \end{bmatrix}, \begin{bmatrix} -3 & 4 \\ 10 & 4 \end{bmatrix} \right\}$$

and

$$\boxed{\dim = 4}.$$

Key Learning Point.

A set of matrices can be tested for linear independence by converting each matrix into its coefficient vector. A non-zero determinant then immediately confirms that all four matrices form a basis.

Question 4

Question.

The vector space spanned by

$$\{3 + 6x + 2x^2 - 10x^3, -2 - 4x - 2x^2 + 10x^3, -3 - 6x - 3x^2 + 9x^3, -2 - 4x + 4x^3\}$$

is a subspace of polynomials of degree $n \leq 3$.

Identify a basis of the subspace and determine its dimension.

Concepts / properties applied:

Coefficient-vector representation; linear dependence; row reduction; pivot columns; basis; dimension.

Method.

Represent the polynomials by their coefficient vectors in the order $(1, x, x^2, x^3)$, row-reduce the coefficient matrix, and use the pivot columns to select a basis from the original polynomials.

Step-by-step working

Let

$$p_1 = 3 + 6x + 2x^2 - 10x^3,$$

$$p_2 = -2 - 4x - 2x^2 + 10x^3,$$

$$p_3 = -3 - 6x - 3x^2 + 9x^3,$$

$$p_4 = -2 - 4x + 4x^3.$$

Their coefficient vectors are

$$v_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \\ -10 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -2 \\ -4 \\ -2 \\ 10 \end{bmatrix},$$

$$v_3 = \begin{bmatrix} -3 \\ -6 \\ -3 \\ 9 \end{bmatrix}, \quad v_4 = \begin{bmatrix} -2 \\ -4 \\ 0 \\ 4 \end{bmatrix}.$$

Form the coefficient matrix using these vectors as columns:

$$C = \begin{bmatrix} 3 & -2 & -3 & -2 \\ 6 & -4 & -6 & -4 \\ 2 & -2 & -3 & 0 \\ -10 & 10 & 9 & 4 \end{bmatrix}.$$

Row-reducing,

$$C \sim \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (\text{Gaussian elimination})$$

Therefore,

$$\boxed{\text{rank}(C) = 3}.$$

The pivot columns are columns 1, 2, 3, so the corresponding original polynomials p_1, p_2, p_3 are linearly independent and form a basis.

Equivalently, the fourth polynomial is dependent on these three. From the RREF,

$$v_4 = -2v_1 - v_2 - \frac{2}{3}v_3. \quad (\text{Pivot/free-column relation})$$

Hence,

$$p_4 = -2p_1 - p_2 - \frac{2}{3}p_3.$$

Verification

Substituting the three selected basis polynomials,

$$\begin{aligned} -2p_1 - p_2 - \frac{2}{3}p_3 &= -2(3 + 6x + 2x^2 - 10x^3) \\ &\quad - (-2 - 4x - 2x^2 + 10x^3) \\ &\quad - \frac{2}{3}(-3 - 6x - 3x^2 + 9x^3) \\ &= -2 - 4x + 4x^3 \\ &= p_4. \end{aligned}$$

Thus p_4 is redundant, while p_1, p_2, p_3 are independent.

Final answer

A valid basis is

$$\boxed{\mathcal{B} = \{3 + 6x + 2x^2 - 10x^3, -2 - 4x - 2x^2 + 10x^3, -3 - 6x - 3x^2 + 9x^3\}}$$

with coefficient matrix

$$\begin{bmatrix} 3 & 6 & 2 & -10 \\ -2 & -4 & -2 & 10 \\ -3 & -6 & -3 & 9 \end{bmatrix}$$

and

$$\boxed{\dim = 3}.$$

Key Learning Point.

Polynomial basis questions can be handled exactly like ordinary vector basis questions by working with coefficient vectors. The rank identifies the dimension, while the pivot columns identify an independent subset of the original polynomials.

Question 5

Question.

The matrices

$$A = \begin{bmatrix} 1 & -2 & -2 & 1 & 1 \\ 2 & -4 & -4 & -2 & 1 \\ 2 & -4 & -4 & 2 & 0 \\ 2 & -4 & -4 & -2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

are related by elementary row operations.

Let

$$W = \text{span} \{1 + 2x + 2x^2 + 2x^3, -2 - 4x - 4x^2 - 4x^3, -2 - 4x - 4x^2 - 4x^3\}.$$

(a)

Determine whether the set is linearly independent. If not, write the second polynomial as a linear combination of the others.

(b)

Find a subset forming a basis for W , and hence determine $\dim W$.

(c)

Find a basis for the row space of A .

Concepts / properties applied:

Linear dependence; scalar multiples; basis; dimension; elementary row operations; row-space invariance.

Part (a) – Linear dependence

Let

$$p_1 = 1 + 2x + 2x^2 + 2x^3,$$

$$p_2 = -2 - 4x - 4x^2 - 4x^3,$$

$$p_3 = -2 - 4x - 4x^2 - 4x^3.$$

Compare p_2 with p_1 :

$$-2(1 + 2x + 2x^2 + 2x^3) = -2 - 4x - 4x^2 - 4x^3.$$

Therefore,

$$\boxed{p_2 = -2p_1}. \quad (\text{Scalar multiple})$$

Since p_3 is identical to p_2 ,

$$p_3 = p_2 = -2p_1.$$

Hence the set is linearly dependent.

The question specifically asks for the second polynomial as a linear combination of the first and third:

$$p_2 = -2p_1 + 0p_3.$$

Therefore the two required coefficients are

$$-2, \quad 0.$$

Part (b) – Basis and dimension

Since

$$p_2 = -2p_1, \quad p_3 = -2p_1,$$

both p_2 and p_3 are redundant.

Therefore,

$$\mathcal{B}_W = \{p_1\}$$

is a basis for W .

Thus,

$$\dim W = 1.$$

Part (c) – Basis for the row space

Since A and B are related by elementary row operations,

$$\text{row}(A) = \text{row}(B). \quad (\text{Elementary row-operation invariance})$$

The non-zero rows of B are

$$(1, -2, -2, 0, 0),$$

$$(0, 0, 0, 1, 0),$$

$$(0, 0, 0, 0, 1).$$

These therefore form a basis for the row space of A .

Verification

There are three non-zero rows in the echelon-form matrix B , and each contains a leading entry in a different column. Hence the three rows are linearly independent.

Therefore,

$$\mathcal{B}_{\text{row}(A)} = \{(1, -2, -2, 0, 0), (0, 0, 0, 1, 0), (0, 0, 0, 0, 1)\}.$$

Final answer

For part (a),

$$\begin{bmatrix} -2, & 0 \end{bmatrix}.$$

For part (b),

$$\mathcal{B}_W = \{1 + 2x + 2x^2 + 2x^3\},$$

$$\dim W = 1.$$

For part (c),

$$\mathcal{B}_{\text{row}(A)} = \{(1, -2, -2, 0, 0), (0, 0, 0, 1, 0), (0, 0, 0, 0, 1)\}.$$

Equivalently, the row-basis matrix is

$$\begin{bmatrix} 1 & -2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Key Learning Point.

A basis retains only the independent directions in a spanning set. Here the polynomial space has dimension one because all three given polynomials lie on the same scalar direction, while the non-zero rows of a row-equivalent matrix provide a basis for the original row space.

Question 6

Question.

The matrices

$$A = \begin{bmatrix} 1 & -1 & -2 & 1 & 1 \\ 2 & -2 & -4 & -2 & 1 \\ 2 & -2 & -4 & 1 & -1 \\ 1 & -1 & -2 & -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

are related by elementary row operations.

Determine whether a subset of the following matrices can form a basis for $M_{2,2}(\mathbb{R})$:

$$M_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad M_2 = \begin{bmatrix} -1 & -2 \\ -2 & -1 \end{bmatrix},$$

$$M_3 = \begin{bmatrix} -2 & -4 \\ -4 & -2 \end{bmatrix}, \quad M_4 = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix},$$

$$M_5 = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}.$$

Concepts / properties applied:

Dimension of $M_{2,2}(\mathbb{R})$; scalar multiples; linear dependence; span; basis.

Method.

Since $M_{2,2}(\mathbb{R})$ has dimension 4, a spanning basis must contain four linearly independent matrices. Check whether the given matrices can supply four independent directions.

Step-by-step working

First observe that

$$M_2 = \begin{bmatrix} -1 & -2 \\ -2 & -1 \end{bmatrix} = - \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = -M_1. \quad (\text{Scalar multiple})$$

Also,

$$M_3 = \begin{bmatrix} -2 & -4 \\ -4 & -2 \end{bmatrix} = -2 \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = -2M_1. \quad (\text{Scalar multiple})$$

Thus,

$$\text{span}\{M_1, M_2, M_3\} = \text{span}\{M_1\}.$$

So the first three matrices provide only one independent direction.

The remaining matrices M_4 and M_5 can contribute at most two additional independent directions.

Therefore,

$$\dim \text{span}\{M_1, M_2, M_3, M_4, M_5\} \leq 1 + 2 = 3.$$

But

$$\dim M_{2,2}(\mathbb{R}) = 4. \quad (\text{Dimension of matrix space})$$

Hence,

$$\text{span}\{M_1, M_2, M_3, M_4, M_5\} \neq M_{2,2}(\mathbb{R}).$$

Therefore no subset of these matrices can form a basis spanning the whole space.

Verification

A basis for $M_{2,2}(\mathbb{R})$ must contain four linearly independent matrices. However,

$$M_2 = -M_1, \quad M_3 = -2M_1,$$

so three of the five candidates collapse to a single independent direction.

Even if both M_4 and M_5 are independent of M_1 and of one another, the largest possible independent subset has size only 3.

Thus four independent matrices cannot be obtained from the supplied candidates.

Final answer

Not possible

Key Learning Point.

To span a 4-dimensional matrix space, four independent directions are essential. Detecting scalar-multiple matrices early can reveal that a proposed spanning set is too small in dimension, making a basis impossible.